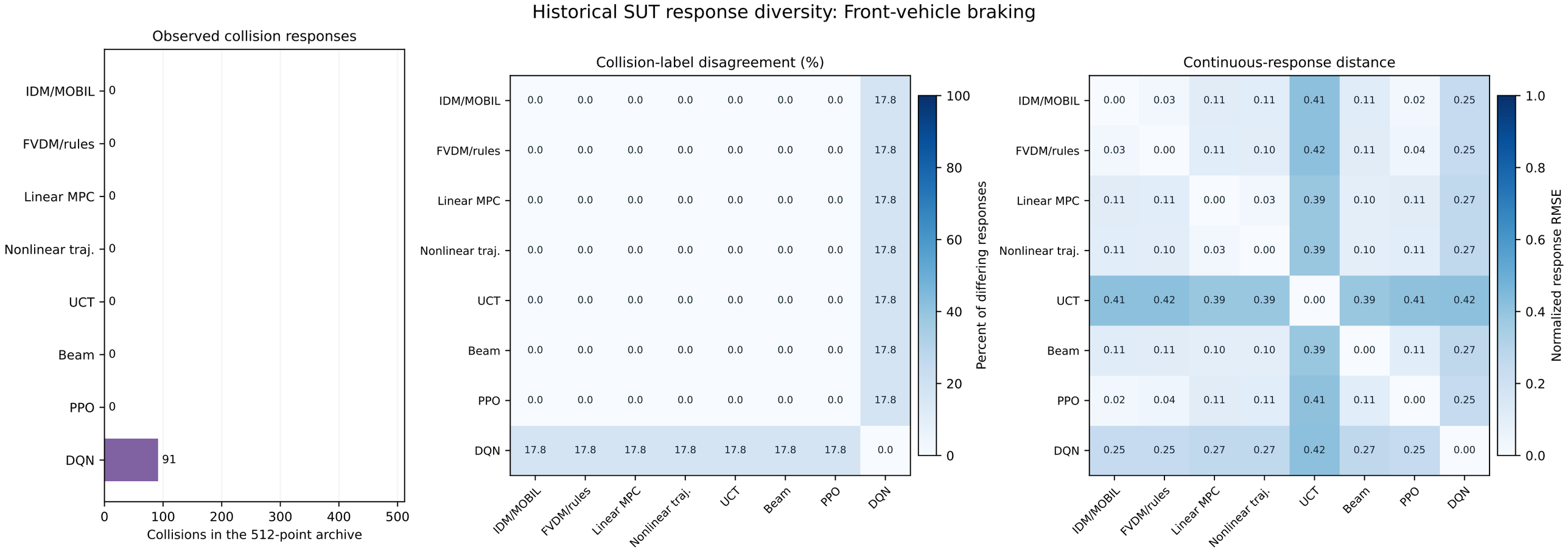
Historical SUT response diversity: Front-vehicle braking
Observed collision responses
IDM/MOBIL
FVDM/rules
Linear MPC
Nonlinear traj.
UCT
Beam
PPO
DQN
0
0
0
0
0
0
0
91
0
100
200
300
400
500
Collisions in the 512-point archive
Collision-label disagreement (%)
0.0 0.0 0.0 0.0 0.0 0.0 0.0 17.8
0.0 0.0 0.0 0.0 0.0 0.0 0.0 17.8
0.0 0.0 0.0 0.0 0.0 0.0 0.0 17.8
0.0 0.0 0.0 0.0 0.0 0.0 0.0 17.8
0.0 0.0 0.0 0.0 0.0 0.0 0.0 17.8
0.0 0.0 0.0 0.0 0.0 0.0 0.0 17.8
0.0 0.0 0.0 0.0 0.0 0.0 0.0 17.8
17.8 17.8 17.8 17.8 17.8 17.8 17.8 0.0
IDM/MOBIL
FVDM/rules
Linear MPC
Nonlinear traj.
UCT
Beam
PPO
DQN
Percent of differing responses
0
20
40
60
80
100
Continuous-response distance
0.00 0.03 0.11 0.11 0.41 0.11 0.02 0.25
0.03 0.00 0.11 0.10 0.42 0.11 0.04 0.25
0.11 0.11 0.00 0.03 0.39 0.10 0.11 0.27
0.11 0.10 0.03 0.00 0.39 0.10 0.11 0.27
0.41 0.42 0.39 0.39 0.00 0.39 0.41 0.42
0.11 0.11 0.10 0.10 0.39 0.00 0.11 0.27
0.02 0.04 0.11 0.11 0.41 0.11 0.00 0.25
0.25 0.25 0.27 0.27 0.42 0.27 0.25 0.00
IDM/MOBIL
FVDM/rules
Linear MPC
Nonlinear traj.
UCT
Beam
PPO
DQN
Normalized response RMSE
0.0
0.2
0.4
0.6
0.8
1.0